

- 26** The scientist Erwin Chargaff analysed the base composition of the DNA of several species. The table shows some of Chargaff's data.

species	percentage of base in DNA			
	adenine	guanine	cytosine	thymine
<i>E.coli</i>	24.7	26.0	25.7	23.6
maize	26.8	22.8	23.2	27.2
grasshopper	29.3	20.5	20.7	29.3
sea urchin	32.8	17.7	17.3	32.1
octopus	33.2	17.6	17.6	31.6
rat	28.6	21.4	20.5	28.4

Which statements can be concluded using **only** the data in the table?

- 1 Cytosine binds to guanine using three hydrogen bonds.
- 2 Approximately 50% of the bases in any of the species are either adenine or thymine.
- 3 The base composition of DNA varies between different species.
- 4 The number of cytosine bases in any of the species is approximately the same as the number of guanine bases.

A 1, 2 and 3 **B** 1 and 4 **C** 2, 3 and 4 **D** 3 and 4 only